Multivariate Calculus Lecture Notes
Section 14.5 – The Gradient and Directional Derivatives

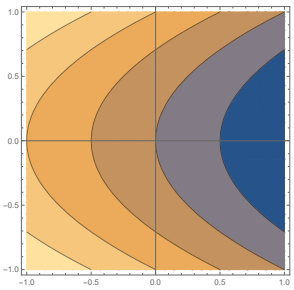
I. Gradients
   a. The *gradient* of a function $f(x, y)$ at a point $P = (a, b)$ is the vector of partial derivatives of f evaluated at P:

$$\nabla f_P = \langle f_x(a, b), f_y(a, b) \rangle.$$

   b. Example: Find the gradient of $f(x, y) = xe^y$ at the point (2,0).
      i. $f_x(x, y) = e^y$; $f_y(x, y) = xe^y$
      ii. Thus the gradient of f is $\nabla f(x, y) = e^y\mathbf{i} + xe^y\mathbf{j}$.
      iii. And $\nabla f(2, 0) = \mathbf{i} + 2\mathbf{j}$.
   c. Algebraic and calculus properties of the gradient: If f and g are differentiable and c is constant then
      i. $\nabla(f + g) = \nabla f + \nabla g$
      ii. $\nabla(cf) = c\nabla f$
      iii. $\nabla(fg) = \nabla f(g) + f\nabla g$
      iv. Chain rule for gradients: If F is a differentiable function of one variable then $\nabla F(f(x, y)) = F'(f(x, y))\nabla f$.

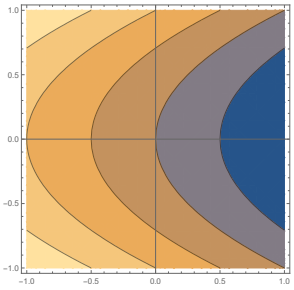
II. Chain rule for paths
   a. Idea
      i. The chain rule for gradients involves stuffing a multivariate function into a single-variable function and finding the gradient.
      ii. There's a chain rule for paths that involves stuffing a single-variable function into a multivariate one and finding the derivative.
      iii. Example
         1. If $\mathbf{c}(t) = \langle x(t), y(t), z(t) \rangle$ gives the position of an object at time t then $T(\mathbf{c}(t))$ could give the temperature of the object at time t.
         2. And $\dfrac{d}{dt}T(\mathbf{c}(t))$ would give the rate of change of the temperature of the object with respect to time.
   b. If f and $\mathbf{c}(t)$ are differentiable, then $\dfrac{d}{dt}f(\mathbf{c}(t)) = \nabla f_{\mathbf{c}(t)} \cdot \mathbf{c}'(t)$.
   c. Example (#19): Calculate $\dfrac{d}{dt}f(\mathbf{c}(t))$, where $f(x, y, z) = xyz^{-1}$ and $\mathbf{c}(t) = \langle e^t, t, t^2 \rangle$ at the point $t = 1$.
      i. First, $\mathbf{c}(1) = \langle e, 1, 1 \rangle$.
      ii. Then $\nabla f(x, y, z) = \langle yz^{-1}, xz^{-1}, -xyz^{-2} \rangle$.
      iii. Next $\nabla f_{\mathbf{c}(1)} = \langle 1, e, -e \rangle$.
      iv. Penultimately, $\mathbf{c}'(t) = \langle e^t, 1, 2t \rangle$, and $\mathbf{c}'(1) = \langle e, 1, 2 \rangle$.
      v. Thus the answer is $1(e) + e(1) - e(2) = 0$.

III. Directional derivatives in the plane
   a. How would you go about finding a derivative in some direction other than the positive x or positive y direction?
   b. Show contour plot of $z = y^2 - x$
      i. Ask about the signs of the derivatives in various directions for a point on a contour line.



ii. If you're on a contour, which directions give you positive directional derivatives, which directions give you negative directional derivatives, and which directions give you zero directional derivatives?

c. Directional derivative definition
   i. Suppose you're at the point (a, b). What is the derivative of $f(x, y)$ in the direction of the unit vector $\mathbf{u} = h\mathbf{i} + k\mathbf{j}$?
   ii. Graphically:



   iii. This is the same as calculating the derivative of f along the path (i.e., the line) defined by $\mathbf{c}(t) = \langle a + ht, b + kt\rangle$ when $t = 0$.
   iv. We have that the *directional derivative* of f at $P = (a, b)$ in the direction of the unit vector $\mathbf{u} = \langle h, k\rangle$ is

$$D_{\mathbf{u}}f(P) = D_{\mathbf{u}}f(a, b) = \frac{d}{dt}f(\mathbf{c}(t))\bigg|_{t=0} = \lim_{t\to 0}\frac{f(a + th, b + tk) - f(a, b)}{t}.$$

   v. This formula is a generalization of the formula for partial derivatives.
      1. If $\mathbf{u} = \mathbf{i}$ then $D_{\mathbf{u}}f(a, b) = f_x(a, b)$, as should be the case.
      2. Similarly, if $\mathbf{u} = \mathbf{j}$ then $D_{\mathbf{u}}f(a, b) = f_y(a, b)$.

d. A shortcut for finding directional derivatives
   i. The definition of $D_{\mathbf{u}}f$ involves the limit of a difference quotient.
      - As with single-variable calculus, there is a shortcut for evaluating directional derivatives.
   ii. Again, $D_{\mathbf{u}}f(a, b) = \dfrac{d}{dt}f(\mathbf{c}(t))\bigg|_{t=0}$, where $\mathbf{c}(t) = \langle a+th, b+tk\rangle$ is a straight-line path in the direction of the unit vector $\mathbf{u} = \langle h, k\rangle$.
   iii. The chain rule for paths says that $\dfrac{d}{dt}f(\mathbf{c}(t))\bigg|_{t=0} = \nabla f_{\mathbf{c}(0)} \cdot \mathbf{c}'(0)$.
   iv. Since $\mathbf{c}(0) = \langle a, b\rangle$ and $\mathbf{c}'(0) = \langle h, k\rangle = \mathbf{u}$, we have the shortcut formula

$$D_{\mathbf{u}}f(a, b) = \nabla f_{(a,b)} \cdot \mathbf{u}.$$

v. If we have a nonzero vector $\mathbf{v}$ that is not necessarily a unit vector, we can find a unit vector $\mathbf{u}$ in the direction of $\mathbf{v}$ via

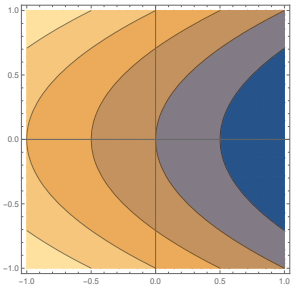
$$\mathbf{u} = \frac{\mathbf{v}}{||\mathbf{v}||}.$$

e. A geometric viewpoint
   i. Another way to view the directional derivative formula is that at the point (a, b), we have $D_{\mathbf{u}}f(a, b) = f_x(a, b)h + f_y(a, b)k$.
   ii. Thus the directional derivative is a weighted (linear) average of the derivatives in the x and y directions.
   iii. Geometrically, this is saying that differentiability of f at (a, b) means that the tangent plane formed by the tangent lines in the x and y directions at (a, b) contains the tangent lines in all of the other directions, too.

f. Example: Find the directional derivative of $f(x, y) = \cos(x+y)$ at the point $(\pi/2, 0)$ in the direction of the point $(0, \pi)$.
   i. Need direction vector: $\mathbf{v} = \langle 0, \pi\rangle - \langle \pi/2, 0\rangle = \langle -\pi/2, \pi\rangle$.
   ii. Normalize to get a unit vector
      1. Magnitude: $\sqrt{\frac{\pi^2}{4} + \pi^2} = \pi\sqrt{\frac{5}{4}} = \frac{\pi}{2}\sqrt{5}$
      2. Unit vector: $\mathbf{u} = \frac{\langle -\pi/2, \pi\rangle}{\pi\sqrt{5}/2} = \left\langle \frac{-1}{\sqrt{5}}, \frac{2}{\sqrt{5}} \right\rangle$
   iii. Partial derivatives: $f_x = f_y = -\sin(x + y)$
   iv. Thus $D_{\mathbf{u}}f(x, y) = -\sin(x + y)\left(-\frac{1}{\sqrt{5}}\right) - \sin(x + y)\left(\frac{2}{\sqrt{5}}\right) = -\frac{\sin(x + y)}{\sqrt{5}}$.
   v. Therefore, $D_{\mathbf{u}}f(\pi/2, 0) = -\frac{1}{\sqrt{5}}$.

IV. More on the gradient

a. Geometric properties of the gradient vector
   i. We have $D_{\mathbf{u}}f(P) = \nabla f_P \cdot \mathbf{u}$.
   ii. Therefore, $D_{\mathbf{u}}f(P) = \|\nabla f_P\|\|\mathbf{u}\| \cos\theta = \|\nabla f_P\| \cos\theta$, where θ is the angle between ∇f_P and $\mathbf{u}$.
   iii. The maximum possible value of $D_{\mathbf{u}}f(P)$ is thus $\|\nabla f_P\|$, since $\cos\theta \leq 1$.
      1. Also, $\cos\theta = 1$ when $\theta = 0$.
      2. Thus $D_{\mathbf{u}}f(P)$ achieves its maximum when $\mathbf{u}$ is in the direction of ∇f_P.
      3. This means that the gradient actually points in the direction of maximum increase of the function.
      4. Graphically:



   iv. Similarly, the minimum possible value of $D_{\mathbf{u}}f$ is $-\|\nabla f_P\|$, since $-1 \leq \cos\theta$.

1. Since $\cos\theta = -1$ when $\theta = \pi$, $D_{\mathbf{u}}f(P)$ achieves its minimum when $\mathbf{u}$ is in the opposite direction of ∇f_P.
2. So the negative of the gradient points in the direction of maximum decrease of the function.

v. What direction is the direction of maximum increase of f at P?
1. Show on the weather map contour diagram.
2. ∇f_P is orthogonal to the level curve of f through P.

vi. Summary: If f is differentiable at P then
1. ∇f_P points in the direction in which f increases the fastest.
2. $-\nabla f_P$ points in the direction in which f decreases the fastest.
3. $\|\nabla f_P\|$ is the maximum rate of change of f at P.
4. $-\|\nabla f_P\|$ is the minimum (most negative) rate of change of f at P.
5. ∇f_P is orthogonal to the level curve of f through P.

vii. What if $\nabla f_P = 0$?
1. This means $D_{\mathbf{u}}f(P) = 0$ for all possible directions.
2. Thus the plane tangent to the graph of f at P is horizontal.
3. So P is probably located at either a relative maximum or a relative minimum of f.

b. Thought experiment: If you're climbing a mountain, and you're at a point P, ∇f_P points in the direction the ascent at P is steepest.

c. Example: Find the direction and value of maximum increase of the function $f(x, y) = x^2 + 2y^2$ at the point (1,2).

i. $\nabla f(x, y) = 2x\mathbf{i} + 4y\mathbf{j}$.

ii. So $\nabla f(1, 2) = 2\mathbf{i} + 8\mathbf{j}$, the direction of maximum increase.

iii. The value of the maximum increase is $\sqrt{4 + 64} = \sqrt{68} = 2\sqrt{17} \approx 8.25$.

V. Functions of three or more variables

a. All of this works when you have three or more input variables, too.

b. There's an interesting application of this with respect to 3D surfaces that satisfy equations of the form $F(x, y, z) = k$, such as the quadric surfaces.

i. Suppose we want to find the equation of the plane tangent to the graph of $\dfrac{x^2}{4} + y^2 + \dfrac{z^2}{9} = 3$ at the point $(-2, 1, -3)$.

ii. This does not give z as a function of x and y, so our previous method for finding tangent planes does not apply.

iii. Remember that to find the equation of a tangent plane in general we need a point in the plane and a vector normal to the plane.

iv. A point is given: $(-2, 1, -3)$.

v. To get a normal vector, note that this surface is the level surface of the function $w = F(x, y, z) = \dfrac{x^2}{4} + y^2 + \dfrac{z^2}{9}$ when $w = 3$.

vi. Thus a vector normal to this surface at $P = (-2, 1, -3)$ is ∇F_P.

vii. $\nabla F(x, y, z) = \dfrac{x}{2}\mathbf{i} + 2y\mathbf{j} + \dfrac{2z}{9}\mathbf{k}$.

viii. Thus $\nabla F(-2, 1, -3) = -\mathbf{i} + 2\mathbf{j} - \dfrac{2}{3}\mathbf{k}$.

ix. Therefore, the equation of the tangent plane is

$$-1(x + 2) + 2(y - 1) - \frac{2(z + 3)}{3} = 0.$$

Homework: Section 14.5 #3, 9, 16, 25, 27, 28, 33, 43, 45, 46